

CORONARY ULTRA CARE SUIT

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**STEM HIGH SCHOOL
FOR BOYS - 6TH OF
OCTOBER**

**GRADE 12,
SEMESTER 1**

**KEY WORDS: CARDIOVASCULAR - NON-INVASIVE
- MONITORING SYSTEM - INFRARED RAYS**

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Abstract

On the track of prospering and developing, Egypt requires tackling **grand challenges** hindering its progress. Public health; proliferation of industrial bases; and improving the scientific and technological environment, are the challenges addressed in this project. **The purpose of the study** is to facilitate the lives of people with Coronary Artery Disease (CAD). Accordingly, a mobile, smart Life Vest was the chosen solution to satisfy our desires. Notably, a manually-fabricated gas pump will dose an anti-coma gas, as a precaution from any sudden glucose fluctuation. This project was **tested** on the design requirements: response time, cholesterol level, glucose level, heartbeats, and oxygen rates. The **results** were organized, and the prototype showed a high capability to maintain a sufficient, safe, and integrated environment for patients. The **major finding** was that if the project is implemented widely by the government, it would significantly reduce the suffering of people with chronic diseases. It was **concluded** that the prototype proposed and successfully controlled an innovative mechanism for achieving the preset **design requirements**, and presenting them through a Mobile Application friendly interface.

Introduction

Egypt's ongoing development efforts are hampered by three major **grand challenges**: public health, industrial base expansion, and the scientific and technological advancement. Thrice of them resulted in the low medical advancements in response to the public health sector's needs. CAD falls within the public health domain, significantly impacting individual well-being and national productivity. **The specific problem** is the Lack of facilitations for coronary artery diseases, and Fig. 1 provides detailed information about the most recent mortality numbers in Egypt, which illustrates an aspect of the problem; the

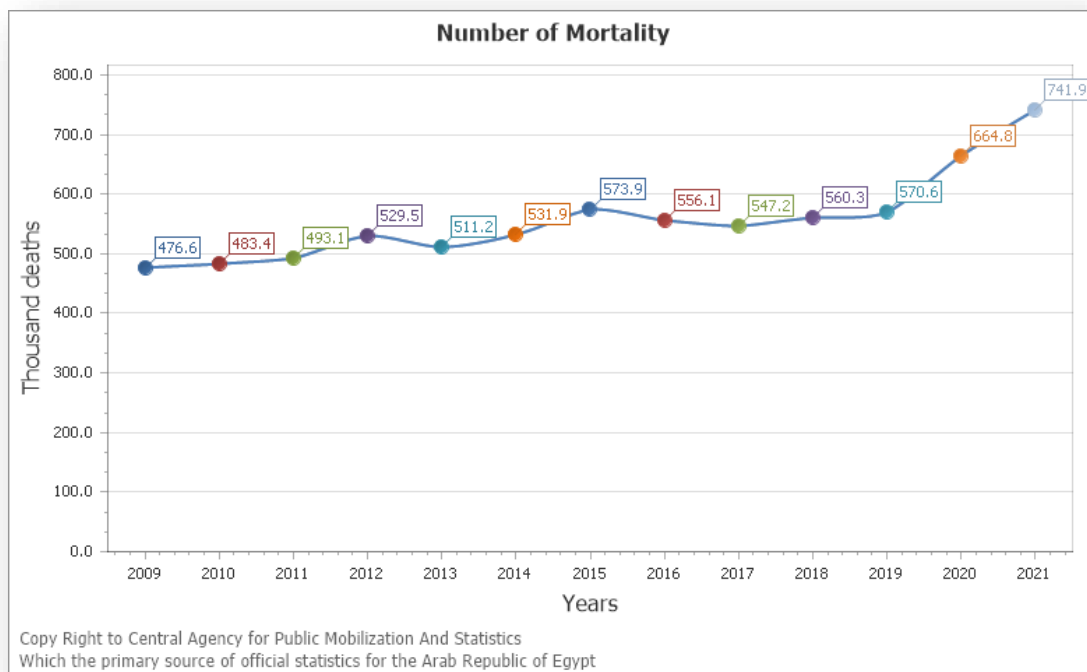


Fig. 1: Most recent mortality numbers in Egypt (2009-2021)

capstone challenge is to design and implement a dependable smart life-jacket to facilitate the lives of people with Chronic Artery Disease (CAD), supported with IoT properties to work as a reliable feedback system.

As reliance on technology is a global orientation, many **prior solutions** were explored and evaluated throughout the entire process. For instance, T: Slim X2 Insulin Pump Device, a point of strength is that the system learns individual user

patterns and adapts insulin delivery accordingly, catering to unique needs and preferences. Conversely, a point of weakness is that it may not be compatible with all devices, potentially limiting some users' options. Additionally, Bigfoot Unity™ Diabetes Management System; point of strength is that supports major brands of both long-acting (basal) and rapid-acting (bolus) disposable insulin pens, it offers versatility; unfortunately, a point of weakness is it has a complex maintenance system and some sensor malfunctions.

The solution pursued was a smart life-vest, integrated with reliable sensors, and supported with a user-friendly, multi-use mobile application. It was **tested** to ensure its validity and its ability to achieve the adjusted **design requirements**, which include ensuring a safe range of cholesterol as follows (125mg/DL-200mg/DL), normal fasting glucose to not fluctuate out of the (70mg/DL-100mg/DL) range, and random blood glucose (70mg/DL-140mg/DL). Heartbeats and Oxygen saturation are aligned with the following ranges (60bpm-100bpm) and (95%-100%) respectively. The **results** demonstrated a crossing-edge ability to engage with the patients in a real-time, offering well-oriented connections with health executives. While also providing information to the user using a mobile app, the **solution** was supported by a machine learning model achieving the highest performance. The materials used and the methodology followed are discussed in the following section in detail.

Materials

Name	Quantity	Usage	Image
Arduino UNO	1	System operation & automation	
Jumper Wires	2 packets	Circuit connections.	
Charger Module	1	Lithium batteries charging	
Solar Panels	1	Source of electric energy.	
Breadboard	2	Electric transferer	
Tin	1 packet	Electronics welding	
ECG sensor	1	HR measurement	
Oximeter sensor	1	Oxygen saturation & HR measurement	
Stepper motor	1	Syringe plunger's position controller	
Driver Board	1	Motor controller	
Bluetooth Module	1	Wireless communication	
LEDs	9	Alerts	

Buzzers	1	Alert producer	
DC-DC Booster	1	Low to high voltage booster	
Battery Holder	1	Battery holding	
Resistors (1K & 220 Ω)	3	Current resisting	
5mm IR led Transmitter & receiver	4	IR signals emitter & receiver	
Lithium batteries	3	Electric energy storage	

Table 1: Materials used throughout the project.

Methods

Designing and Visualization:

1. First of all, a 3D model was designed using SketchUp to visualize the project before constructing it as shown in Fig. (2).



Fig.2: 3D-model for the prototype.

2. Secondly, a schematic diagram was made using Fritzing in order to construct the smart system as shown in Fig. (3).

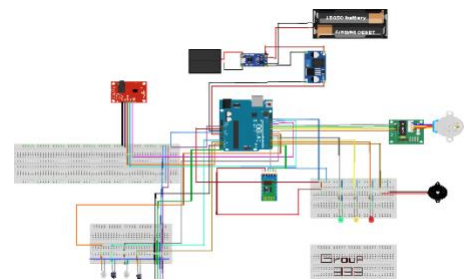


Fig. (3) shows the schematic diagram of the project

Construction Steps:

1. In the first step, the electronic components were made and divided into three sections: the first part is the Oximeter and IR transmitter and receiver LEDs, the second part is the ECG with stepper motor, and the third part is the Bluetooth module with Buzzer and LEDs for alerting and communication the as shown in Fig. (4).
2. In the second step, all electronic components were put in the vest except IR LEDs, and the Oximeter was put in the gloves. Behind the vest, there was a solar panel to provide sustainable charging for all electronic components as shown in Fig. (5).

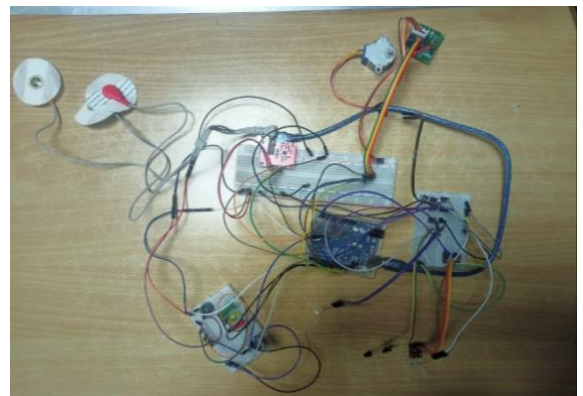


Fig. (4) shows the three electronic parts of the system

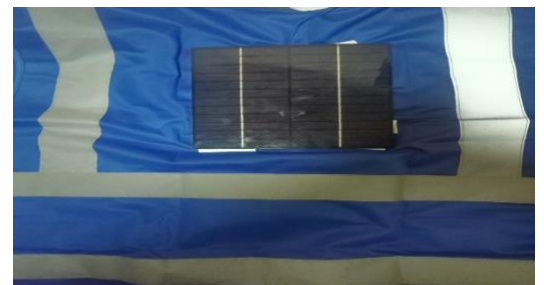


Fig. (5) shows the solar panel position for providing the system with power

3. In the third step, the machine learning model is made using Python programming language with its main libraries via Google Colab which helps in predicting if the user has coronary artery disease or not as shown in Fig. (6).

```
[8] import numpy as np
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score
```

Fig. (6) shows the main libraries used in the machine Learning model

4. In the fourth step after predictions and measurements, a mobile application was built using Flutter via Android Studio to connect with Arduino UNO via Bluetooth show the results, and alert the user.

Design Requirements:

To ensure that the project achieves its required target, four design requirements are chosen to check its performance:

1. The system's response mustn't exceed two minutes.
2. The heartbeats must be tracked and maintained within a range of 60 to 100 beats per minute.
3. The oxygen saturation must be tracked and maintained within a range of 94% to 100%.
4. The glucose and cholesterol concentrations must be tracked and maintained with range (70 mg/dL – 140 mg/dL) and (125 mg/dL – 200 mg/dL) respectively.

Test Plan:

1. The solar panel was subjected at the back of the vest and the power was measured using a multimeter during the day.

- The oximeter and ECG sensors were stuck on the finger and chest respectively; to measure the heartbeats and the oxygen saturation in the blood as shown in Fig.(7a & 7b).



Fig. 7a shows the ECG Connection

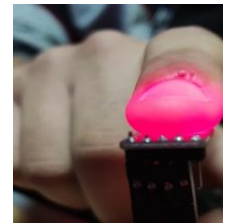


Fig. 7b shows the Oximeter functionality of IR

- IR transmitter and receiver LEDs are stuck by fingertips to measure the cholesterol and glucose levels, as shown in Fig. (8).

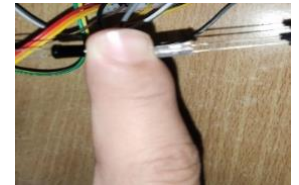


Fig. 8 shows the IR LEDs with fingertips to make its non-invasive measurements.

- After getting the measurements, the coronary artery disease is predicted via a machine learning model (using Python programming language), and its sample code is shown in Fig. (9).

```
input_data = (41, 0, 204, 33, 105, 23.3, 103, 1, 1)

# change the input data to a numpy array
input_data_as_numpy_array= np.asarray(input_data)

# reshape the numpy array as we are predicting for only on instance
input_data_reshaped = input_data_as_numpy_array.reshape(1,-1)

prediction = model.predict(input_data_reshaped)
print(prediction)

if (prediction[0]== 0):
    print('The Person does not have a Coronary Artery Disease')
elif (prediction[0]== 2):
    print('The Person has a Heart Disease')
elif (prediction[0]== 3):
    print('The Person has diabetes')
else:
    print('The Person has Coronary Artery Disease')
```

[1.]
The Person has Coronary Artery Disease

Fig. 9 shows the prediction machine learning sample code.

- The Arduino UNO is connected to the mobile application via Bluetooth and results have appeared in the app as shown in Fig. (5) to facilitate the patient's life and achieve the design requirements.



Fig. 10 shows the results UI via app

6. If the results didn't match the design requirements, an alert is active as shown in Fig. (6), and during the process, the response time was measured.

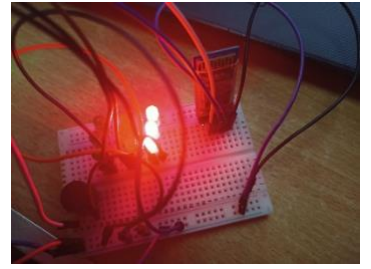


Fig. 11 shows the alert electronic system

Results

Negative results:

The ECG electrodes were placed on the human chest in a wrong pattern, as a result, the resulting graph appeared irregular, deviating from the typical criteria observed in standard ECG graphs.

Positive results:

The results of this project are represented in measuring the heartbeats as shown in Fig. 11, O₂ saturation as shown in Fig. 12, glucose as shown in Fig. 13.

Moreover, the accuracy of measuring the HR and SpO₂ was calculated by comparing the actual results using an oximeter device and the estimated results using the oximeter sensor. Finally, the relation between the power input by the photovoltaic cells and the hour in the day resulting in such output is presented in Fig. 14 and table 2.

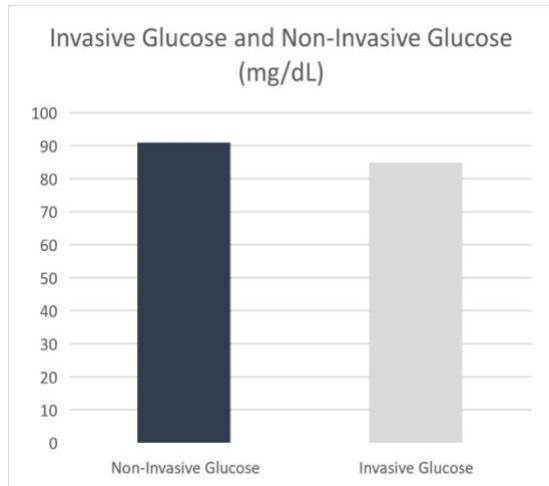


Fig.14: Illustrates the difference between the invasive and non-invasive readings.

TIME	VOLTAGE (V)	INTENSITY (A)	POWER (W)
06:00 am	12.25 ± 0.5%	0.03 ± 0.8%	0.37 ± 1.3%
08:00 am	13.40 ± 0.5%	0.11 ± 0.8%	1.5 ± 1.3%
10:00 am	13.04 ± 0.5%	0.14 ± 0.8%	1.8 ± 1.3%
12:00 pm	13.40 ± 0.5%	0.35 ± 0.8%	4.7 ± 1.3%
02:00 pm	13.42 ± 0.5%	0.37 ± 0.8%	5.0 ± 1.3%
04:00 pm	13.26 ± 0.5%	0.24 ± 0.8%	3.2 ± 1.3%
06:00 pm	12.32 ± 0.5%	0.01 ± 0.8%	0.1 ± 1.3%

Table 2: Voltage-Intensity-Power and time table.

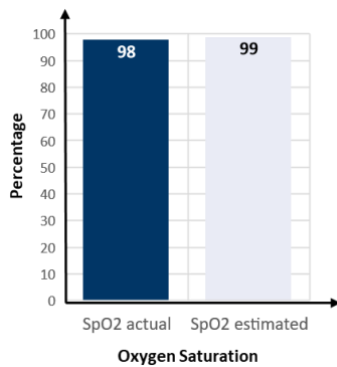


Fig.13: Illustrates the SpO2.

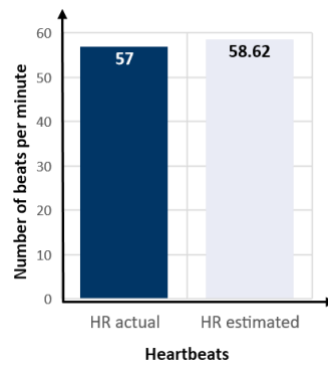


Fig.12: Illustrates the HR records.

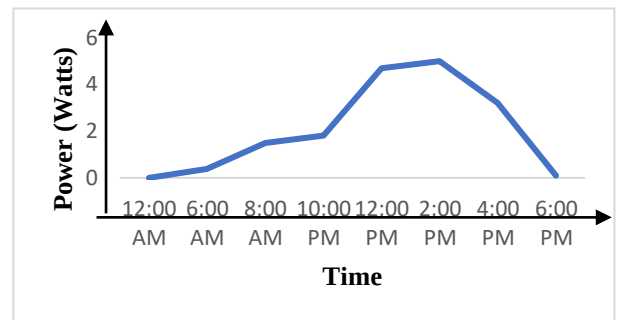


Fig.15: Power and Time graph

ME.1.02 The area under curve of the power-time was measured to be $2,4 \times 10^5$ joules, to supply the system with. In addition, the system's accuracy is 95.7%, it was determined by calculating the percentage error according to the law discussed in **PH.1.01**:

$$\frac{\text{Estimated} - \text{Actual}}{\text{Actual}} * 100 \quad \text{equation (1)}$$

Cholesterol is measured with the same methodology of Glucose non-invasive measurement with different molar absorptivity value, giving 162 mg/dl. Finally, the patient was alerted when any of the readings didn't meet the desired range in 15 second as a response time of our system, achieving the design requirement.

Analysis

Coronary Artery Disease (CAD):

One of the most harmful cardiovascular disorders, is mainly identified by narrowing and blocking of coronary arteries, due to atherosclerosis. Atherosclerosis is a cardiovascular condition marked by fatty deposits and cholesterol accumulation in the walls of the arteries. In addition, it is known as a chronic inflammatory disorder due to the

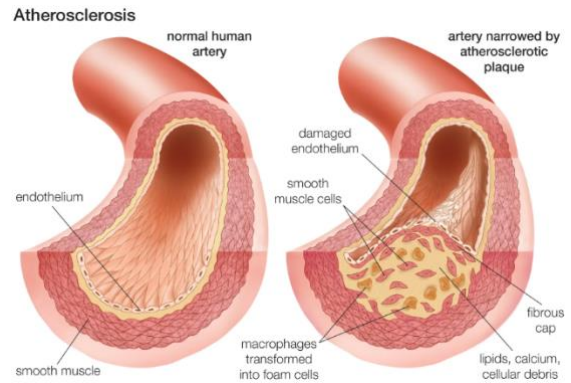


Fig.16: Macrophages cells leakage through the arterial wall.

endothelial cells' dysfunction allowing lipids, inflammatory, and foam (macrophages) cells to leak through the arterial wall forming a plaque as shown in Fig. 16. This leak is mainly occurred due to the Chemotaxis which attracts the immune cells to the site of inflammation or injury by chemical signals called chemokines which are produced by infected cells and guide immune cells towards the target area. Within the chemotaxis, the cell membrane's permeability increases to overcome the damage and the defect of the infected cells by passing immune cells (macrophages) and proteins. As explained in the concept of the cell membrane and its permeability illustrated in **BI.1.02**, the cell usually aims to reach an equilibrium state by changing the cell's permeability.

Secondly, lipids existence is one of the essential components of the cell, if their levels become abnormally high, Hyperlipidemia which refers to the elevated levels of lipids in the blood is caused, including cholesterol (LDL) and triglycerides. Elevated triglyceride levels form small, dense LDL particles, which are more prone to penetrating the arterial wall. The interaction between LDL and immune cells (macrophages) - in the damaged area - contributes to the formation

of foam cells. As a result, chemotaxis takes place, increasing macrophages – due to the immune response- in the infected area, ending with plaque formation.

Furthermore, the increase in glucose levels, heartbeats, and the defect in O₂ saturation - resulting in ischemia - contribute to increasing permeability and endothelial cells' dysfunction. This dysfunction allows the inflammatory cells deposition, lipids infiltration, and foam cells accumulation, ending in plaque formation as shown in Fig.17.

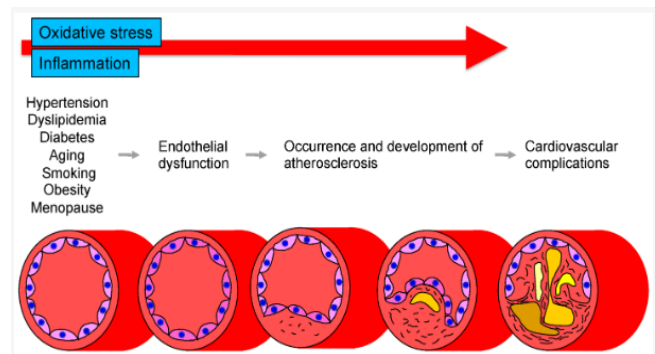


Fig.17: Dysfunction's consequences illustration.

Moreover, diabetes - with its metabolic impact - intensifies atherosclerosis illustrating the role of hyperglycemia in endothelial dysfunction and plaque formation. Elevated blood glucose levels stimulate the liver to produce more glucose through a process known as gluconeogenesis. Increased glucose production can lead to higher levels of triglycerides as the excess glucose is converted into triglycerides for storage. In addition, hypertension increases mechanical stress on arteries which accelerates atherosclerosis.

Glucose & Cholesterol Algorithmic Analysis:

The near-infrared region of the electromagnetic spectrum (from 780 nm to 2500 nm, as shown in Fig.18) is used by near-infrared spectroscopy (NIR). Diagnosing systems for blood sugar and cholesterol are invasive in nature. The diagnosing procedure is performed by drawing blood by puncturing the skin with a needle and

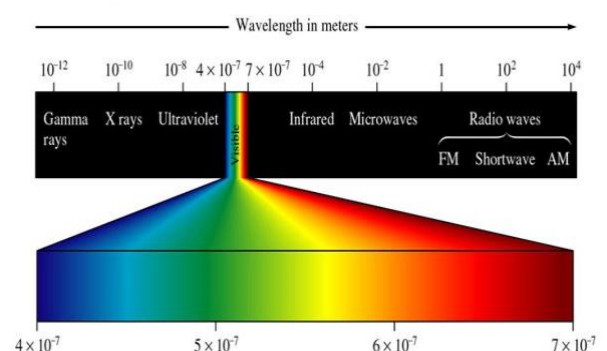


Fig. 18: Light skin penetration with different wavelengths (nm)

applying it to a chemically active test strip. Conducting the tests and calculating their results do not require a supply of blood.

Glucose ($C_6H_{12}O_6$) is a kind of monosaccharide in the form of pyranose. It has several absorption peaks in the NIR region. Glucose has light absorption peaks at wavelengths of 940 nm, 970 nm, 1197 nm, 1408nm, 1536nm, 1688nm, 1925 nm, 2100nm,

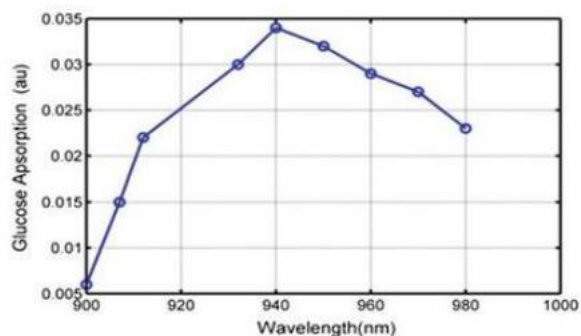


Fig. 19: Shows the highest Glucose's light absorption peak.

2261 nm, and 2326nm, but at the 940 nm wavelength, the attenuation of optical signals by other constituents of the blood, as depicted in Fig.19. When the wavelength hits its optimum, the desired depth of penetration can be then achieved, and the actual glucose concentration can be predicted.

Cholesterol ($C_{27}H_{46}O$) is a sterol molecule that is a crucial lipid present in cell membranes and serves in the synthesis of steroid hormones and vitamin D. In the context of near-infrared spectroscopy (NIR), cholesterol exhibits specific absorption peaks in the same near-infrared region; see Fig.20. Additionally, its absorption peaks in the NIR region occur at numerous wavelengths. These absorption peaks show molecular vibrations and electronic transitions that are only found in cholesterol. This lets NIR spectroscopy find and measure cholesterol in samples.

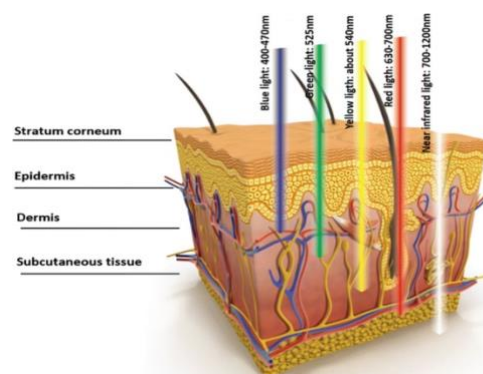


Fig.20: Light skin penetration with different wavelengths (nm)

The Beer-Lambert law, as shown in equation (1), is commonly applied to chemical analysis measurements and especially to non-invasive methods to determine the concentration of glucose and cholesterol that absorb light. In equation (1), A is absorbance, ϵ is the molar absorptivity (which depends on the nature of the chemical and the wavelength of the light used), l is the length of the

path light must travel in the solution in centimetres, and c is the concentration of a given solution.

$$A = \epsilon lc$$

equation (2)

The measurement of the number of photons delivered at a point in each unit of time is called the intensity I .

The intensity of the beam of light, which is measured entering our sample (I_0), is compared with the intensity of the beam of light exiting our sample (I), as Fig.21 depicts.

Then we can take the ratio I/I_0 to get an indication of what fraction of the light entering the sample was found exiting the sample.

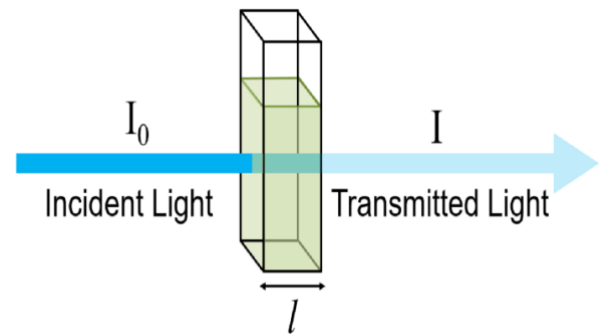


Fig. 21: Illustrates the flow of I and I_0

$$A = -\log\left(\frac{I}{I_0}\right)$$

equation (3)

This formula ($-\log\left(\frac{I}{I_0}\right) = \epsilon lc$) is used and converted to be an algorithm for predicting cholesterol and glucose.

Mobile Application Monitoring System Interface:

CS.2.05 discusses Mobile App development using Java and Android Studio. Studying Mobile Apps gave the inspiration to develop an IoT monitoring system using Google Flutter. Firstly, the Internet of Things (IoT) is the communication between things, or systems, wirelessly to transfer data between each other. Communication occurs through devices, such as Near-field communication (NFCs), Radio-frequency identification (RFIDs), wireless and Bluetooth modules, without any form of human or machine interactions (Li et al., 2014). HC-05 Bluetooth module is used for the IoT connection since it is easy to use and supports Serial Port Protocol, which is supported by Arduino.

A standout feature, the application can expect potential health risks based on the monitored parameters. In critical situations, the app can automatically initiate contact with predefined relatives, providing them with an active location and a detailed report of the patient's recorded data. This feature ensures timely intervention in emergencies.

Secondly, Google Flutter is an open-source cross-platform app development framework developed by Google to work with Dart Programming Language. Cross-platform development gives Flutter an advantage over Java: it can be used to develop Android, IOS, Web Apps, and

```

10 BluetoothConnection.toAddress(widget.server.address).then((connection) {
11   print("Connected to the device");
12   connection = _connection;
13   setState(() {
14     isConnected = true;
15     isDisconnecting = false;
16   });
17 }

```

Fig. 22: Shows the connection between the flutter app and the Arduino code.

more using a single source code. The significance of real-life monitoring lies in ensuring that the system's processes are running smoothly while achieving optimal parameters. Finally, Fig.22 shows an example of Flutter source code, the code that connects the app to the Bluetooth module.

Electrocardiography (ECG) Sensor:

An ECG (electrocardiograph) sensor is designed to measure and record the electrical activity of the heart over a specific period. The resulting information is typically represented as a graph or series of waveforms known as an electrocardiogram. An ECG sensor includes electrodes that are strategically placed on the skin. These electrodes detect the electrical signals generated by the heart's electrical impulses produced as a result of depolarization and repolarization of the heart's chambers.



$$\text{Heart Rate} = \frac{300}{4} = 75 \text{ per minute}$$

Fig. 23: Illustration for the Ideal ECG graph and Calculation.

The resulting waveforms are divided into three main types of waves such as: P-wave indicating the electrical activity as the impulse spreads through the atria,

causing them to depolarize, QRS complex (R-wave) indicating the spread of the electrical impulse through the ventricles, leading to their depolarization, and T-wave indicating the recovery of the ventricles (repolarization) as they prepare for the next contraction.

The resulting graph is then converted into the needed heart pulse readings by using the RR formula that calculates the number of beats per minute from the given crests and troughs of the R-wave in the graph as shown in Fig. 23.

$$\text{HR} = 300 / \text{no. of units in the RR interval}$$

Where 300 here is a constant that refers to the number of units per minute, each unit equals 0.2 seconds, so by dividing the total number of units by the number of units taken in one pulse, the heart rate is determined.

Oximeter Pulse Meter:

A pulse oximeter operates on the photoelectric plethysmography principle. The accuracy of oximeters may fluctuate due to factors like motion artifacts, ambient light, and certain medical conditions: anemia. A pulse oximeter emits light through a pulsatile component of the vascular system. Additionally, the device contains two light-emitting diodes (LEDs).

1. Red spectrum (**660 nm**)
2. Infrared spectrum (**940 nm**).

Deoxygenated hemoglobin absorbs more red light while transmitting more infrared light. It absorbs more in the red range and less in the infrared range. In contrast, oxygenated hemoglobin absorbs more infrared light while allowing more red light to pass through.

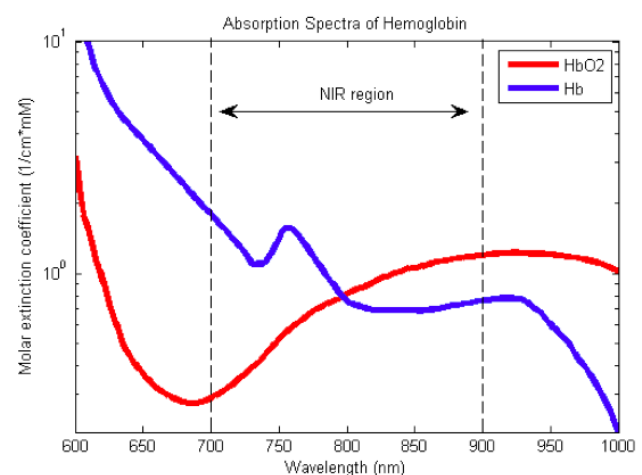


Fig. 24: Oxygenated and Deoxygenated Hemoglobin

It absorbs more in the infrared spectrum and less in the red spectrum [1], as seen in Fig. 24.

In the context of Glucose: glucose tends to absorb more infrared spectrum. This means that when infrared light passes through a blood sample containing glucose, the glucose molecules absorb a significant amount of the infrared light, leading to a higher oximeter's readings.

The blood oxygen content is calculated using the differences in the readings. The amount of red and infrared light absorbed by oxygenated and deoxygenated haemoglobin in the blood is compared.

Less glucose causes more light scattering, shortens path lengths, and reduces tissue uptake, and vice versa. Because of greater absorption in the glucose-rich tissue, the reflected light has a lower intensity than in the glucose-depleted tissue; Fig. 25 illustrates this description.

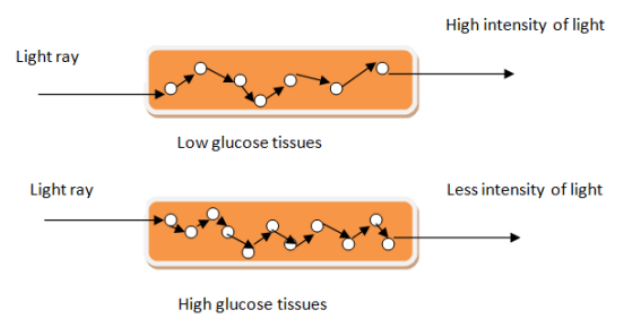


Fig.25: Light Reflection's principles.

Conclusion

Regarding the successful completion of the test plan, an appreciation is required for the effort done to construct and design this project, starting with researching Egypt's grand challenges and the specific problem (the lack of facilitations for coronary artery disease), to finding prior solutions and constructing our own according to specific design requirements, to designing and building our prototype, which achieved great success based on specific scientific bases. The project showed unexpected results and proved its efficiency in monitoring the patient's health (heartbeats, oxygen saturation, glucose, and cholesterol levels) through a wearable smart monitoring system. In addition, the automated solution is capable of alerting the patient if there is any exceed or fall below the optimum ranges of the desired parameters. This project is an opening gate for upcoming research to help cardiovascular chronic diseases (coronary artery disease) carriers by monitoring, alerting, and contributing to their treatment.

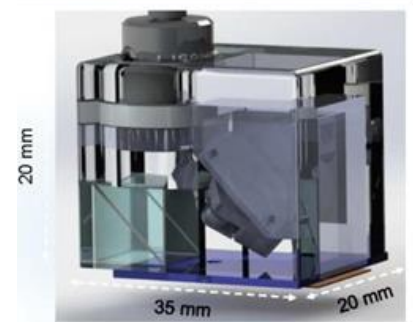
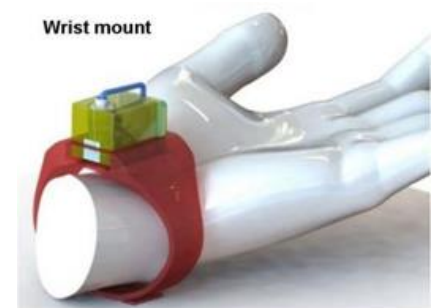
Recommendations

Photoacoustic Imaging for Oxygen Saturation:

For further oxygen saturation calculation accuracy, photoacoustic imaging is an innovative medical approach that combines the principles of acoustics and optics. It's a non-invasive imaging method that is used to visualize structures and function within the body, pursued by the device illustrated in Fig.26. The generation of acoustic waves in tissue from the absorption of pulsed laser light is the very basic principle behind photoacoustic imaging.

Narrowing down the scope to have a glimpse at the working mechanism, it initiates a short-pulsed laser to illuminate tissue. When laser light reaches biological tissues, it is then absorbed by chromophores, as they are molecules that absorb light. Declaring, Hemoglobin is the molecule responsible for carrying oxygen and is a crucial chromophore.

Then, ultrasonic waves or acoustic waves arise in the tissue thanks to a thermal expansion, as a result of rapid heating caused by the absorbed light. Accordingly, ultrasound transducers are considered detectors; the oxygenated and deoxygenated haemoglobin's information is given according to the timing and intensity of the detected signals. Compared to traditional optical imaging, photoacoustic imaging can penetrate deeper into tissues, making it suitable for imaging structures within the body.



Unpublished, Duke Photoacoustic Lab
<https://photoacoustics.pratt.duke.edu/>

Fig.26: 3D-model for the device.

(low-density lipoprotein) cholesterol:

Cholesterol meters, commonly known as cholesterol testing devices or cholesterol monitors, typically measure total cholesterol, high-density lipoprotein (HDL) cholesterol and triglycerides. Some advanced cholesterol meters may also provide an estimate of low-density lipoprotein (LDL) cholesterol, but they exist only in laboratories and are not wearable.

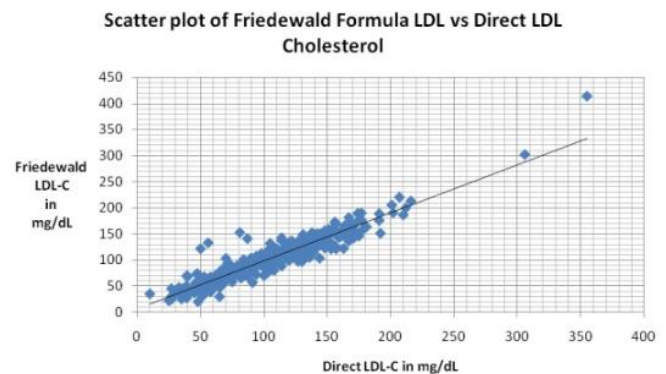


Fig.27: Illustration for the relation between Friedewald Formula and the LDL cholesterol.

LDL cholesterol could be calculated using an efficient Fried Ewald equation, which estimates LDL cholesterol based on measurements of total cholesterol, HDL cholesterol, and triglycerides; also, Fig.27 depicts the relation between the equation and the LDL glucose. The equation is as follows:

$$\text{LDL} = \text{Total cholesterol} - \text{HDL cholesterol} - (\text{Triglycerides}/5)$$

Noninvasive glucose:

A non-invasive glucose monitoring meter offers a convenient and pain-free alternative to conventional methods and more efficient measurement compared to ours. It depends on infrared spectroscopy, electromagnetic sensors, and nanotechnology to analyze glucose levels without penetrating the skin. Moreover, it transmits the data wirelessly to the mobile application that displays the reading as shown in Fig. 28. **IR Spectroscopy** and **electromagnetic Sensors** transmit waves through the skin and receive them back with changes in their properties, which can be indicative of glucose concentration.



Fig.28: Photograph for the device noted.

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